

NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Balkumari, Lalitpur



(Affiliated to Pokhara University)

A Lab Report

On

Subject:- Web Technology.



Lab Report # 1

Title:- Introduction to HTML

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OBJECTIVE:- TO BE FAMILIARIZATIZED WITH THE BASIC STRUCTURE OF HTML AND HTML TAGS.

Theory:

1.Introduction:

Visual Studio Code (often shortened to VS Code) is a free, open-source code editor developed by Microsoft. It's widely used by programmers because it combines simplicity with powerful features. Here's what makes it stand out.

☑Key Features

- Cross-platform: Works on Windows, macOS, and Linux.
- Lightweight but powerful: Faster than full IDEs, yet supports advanced development.
- Built-in support Syntax highlighting, debugging, Git integration, and IntelliSense (smart code completion).
- Extensions marketplace: Thousands of plugins for languages (Python, C++, JavaScript, etc.), frameworks, themes, and tools.
- Extensions marketplace: Thousands of plugins for languages (Python, C++, JavaScript, etc.), frameworks, themes, and tools.
- Customization: Highly configurable with settings, keybindings, and themes.
- Integrated terminal: Lets you run commands without leaving the editor.

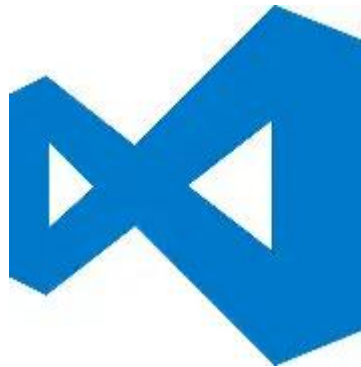


Fig:Visual Studio Code

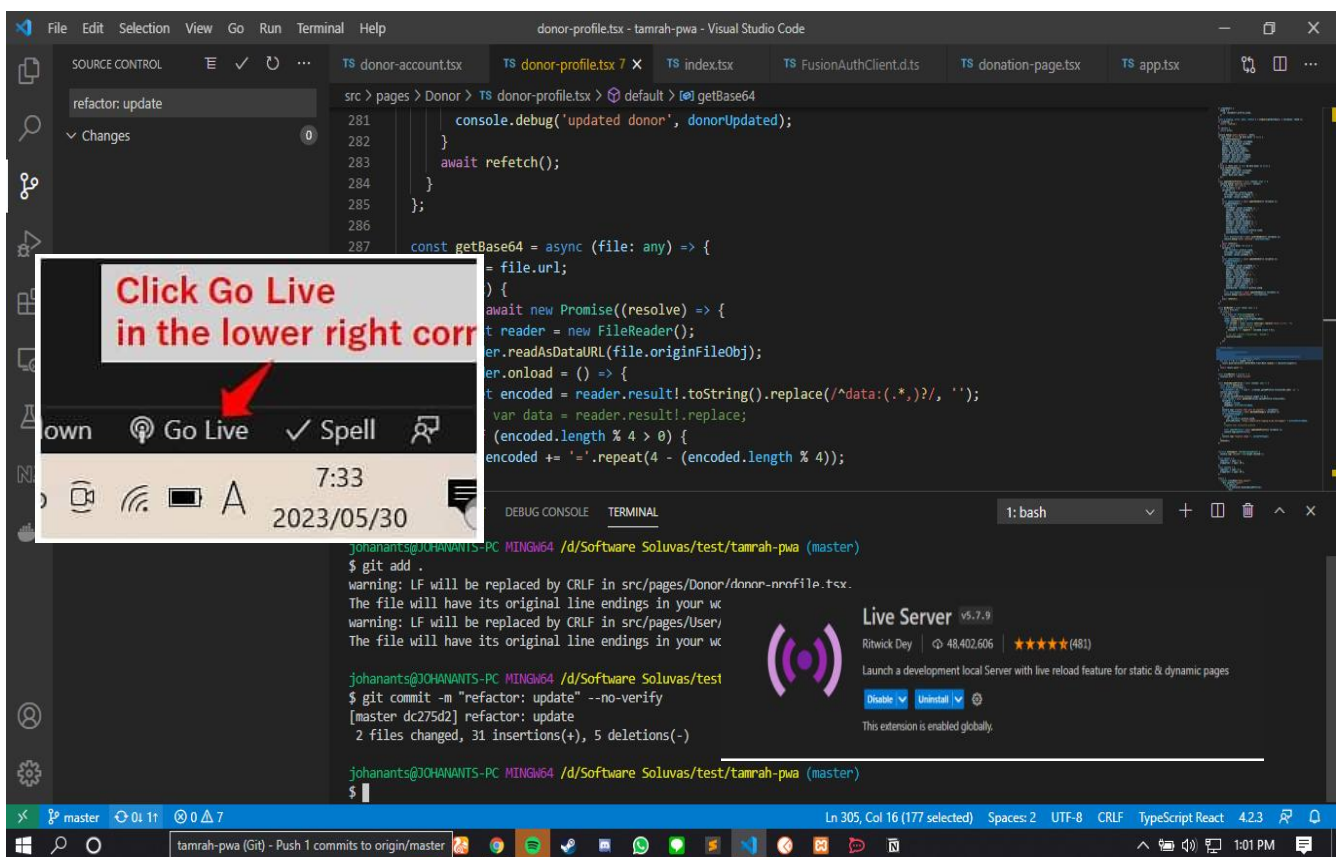


Fig:Live server and Vs code user interface

1.1.Explanation:

The Visual Studio Code (VS Code) interface is a user-friendly environment that allows developers to work on individual files or open folders to create a workspace. It includes a range of tools and features to enhance the development process. The Live Server extension is particularly useful for front-end developers, as it automates the process of creating a local development server, enabling automatic browser refresh upon file changes. This feature significantly enhances productivity and eliminates the need for manual refreshing, making it an essential tool for web development.

1.2.What is live server?

--The live server is popular tool for web developers using visual studio code.It automatically open your web project in a browser and refreshes the page Whenever you save changes to your HTML, CSS, or JavaScript files.This features significantly enhances productivity by eliminating the need for manual browser refresh.

1.3.Steps to Install Live Server in VS Code

1. Open Visual Studio Code Launch the application on your computer.
2. Access the extension panel.
3. Search for live server
4. Install the extension

1.4.How to use Live Server

1. Open your project
2. Open an HTML File
3. Start live server:-
 - Click the Go live button in the bottom right corner of the VS Code window. Alternatively, you can right click on the html file and select Open with Live server.
4. Automatic refresh
5. Stop live server

2. HTML

HTML (**HyperText Markup Language**) is the standard language for creating and structuring web pages using tags and elements. It defines how content like text, images, and links appear in a browser.

- It is a markup language, not a programming language.
- This means it annotates text to define how it is structured and displayed by web browsers.
- It is a static language, meaning it does not inherently provide interactive features but can be combined with CSS for styling and JavaScript for interactivity.

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

Fig:code for HTML

- The <!DOCTYPE html> declaration defines that this document is an HTML5 document
- The <html> element is the root element of an HTML page
- The <head> element contains meta information about the HTML page
- The <title> element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The <body> element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The <h1> element defines a large heading
- The <p> element defines a paragraph.

2.1.HTML Strucutre

HTML Page Structure

Below is a visualization of an HTML page structure:



Fig:HTML page structure

2.2HTML Attributes

- All HTML elements can have **attributes**
- Attributes provide **additional information** about elements
- Attributes are always specified in **the start tag**
- Attributes usually come in name/value pairs like: **name="value"**

2.2HTML Elements and Tags

- HTML Tages are the keywords on a webpage that define how your web browser must format and display your web page.
- It is the fundamental syntactic units of HTML.
- HTML tags are used to hold the HTML element.
- HTML attributes are used to describe the characteristic of an HTML element in detail.

HTML Tags are categorized into 3 classes:

Class1: paired and unpaired

Paired tag is tag having both start tag and end tag.

Unpaired tag have either start tag or end tag.

Class2: inline and block level.

Inline tag is doesnot start on a new line and it only takes up as much width as necessary. Eg:

Block tag starts on a new line and take up the full width of their container.

Class3: physical and logical

Physical tag: These tags are used to provide the visual appearance to the text. Example: , <i></i> , <strike></strike> ,etc.....

Logical tag: These tags are used to add some logical or semantic value to the text. Example: `<strong></strong>` , `<em></em>` , `<del></del>` etc.....

Start tag	Element content	End tag
<code><h1></code>	My First Heading	<code></h1></code>
<code><p></code>	My first paragraph.	<code></p></code>
<code>
</code>	<i>none</i>	<i>none</i>

2.2.1Nested HTML Elements

HTML elements can be nested (this means that elements can contain other elements).

All HTML documents consist of nested HTML elements.

The following example contains four HTML elements (`<html>`, `<body>`, `<h1>` and `<p>`):

Example

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

2.2.2Empty HTML Elements

HTML elements with no content are called empty elements.

The `<br>` tag defines a line break, and is an empty element without a closing tag:

Example

```
<p>This is a <br> paragraph with a line break.</p>
```

2.3HTML Horizontal Rules

The <hr> tag defines a thematic break in an HTML page, and is most often displayed as a horizontal rule.

The <hr> element is used to separate content (or define a change) in an HTML page: The <hr> tag is an empty tag, which means that it has no end tag.

Example

```
<h1>This is heading 1</h1>
<p>This is some text.</p>
<hr>
<h2>This is heading 2</h2>
<p>This is some other text.</p>
<hr>
```

TASK: Heading(h1-h6) and paragraph(p) tags Br and hr tags, superscripts subscripts tag.

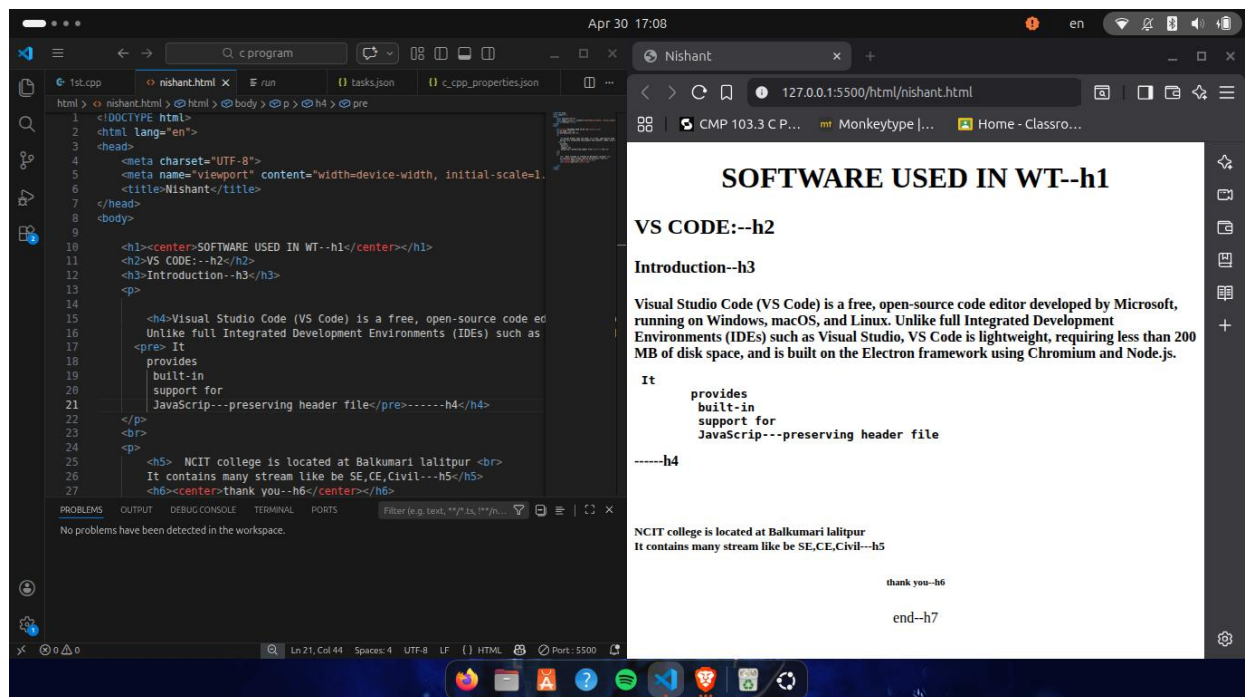


Fig: Heading tags and paragraphs

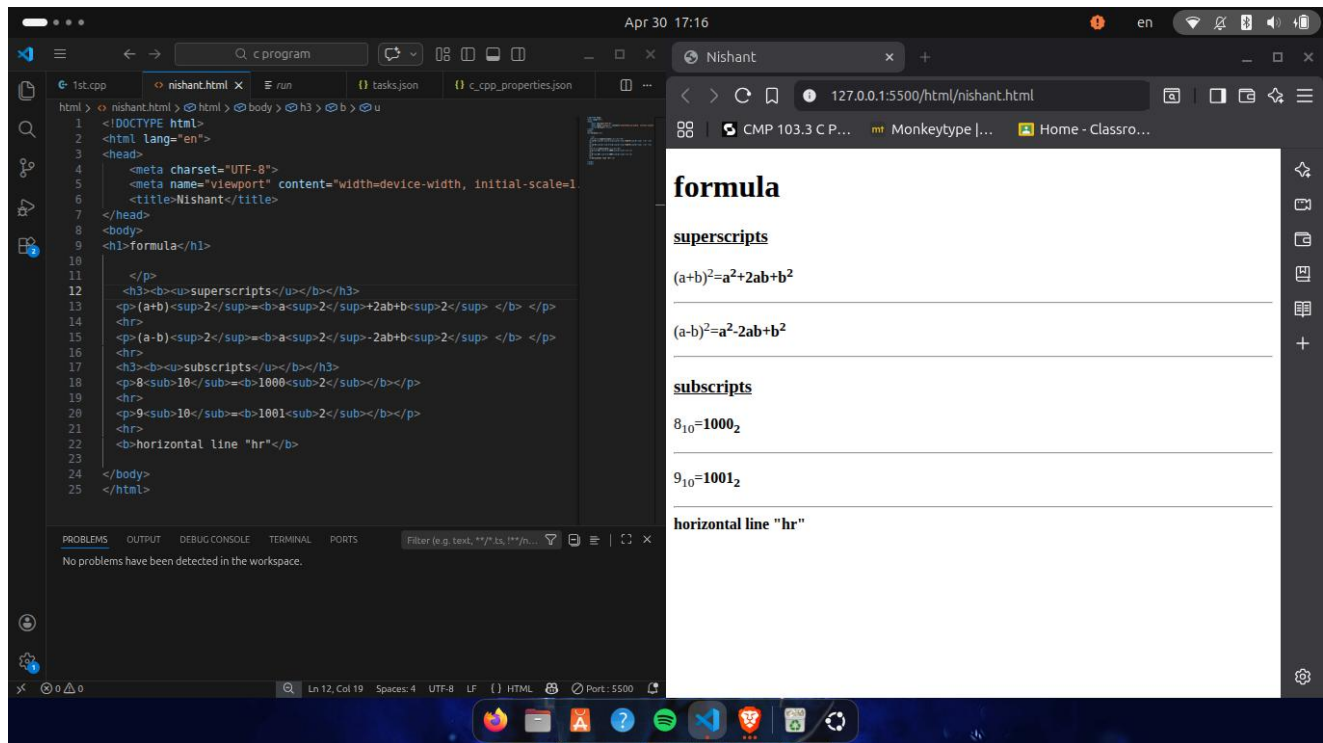


Fig:code of hr,superscripts and subscripts tag

Detailed Tag Explanations with Code

1. <h1> — Main Page Title

The <h1> tag defines the **most important heading** on a page. It should be used **only once** per page for the main topic. Browsers render it as the largest text by default.

```
<h1><center>SOFTWARE USED IN WT--h1</center></h1>
```

2. <h2> — Major Section Heading

The <h2> tag marks **major sections** of a page. Multiple <h2> tags can exist on a single page, one per major topic.

```
<h2>VS CODE:--h2</h2>
```

3. <h3> — Sub-section Heading

The <h3> tag marks a **sub-section** under an <h2>. It creates a third level of hierarchy in the document structure.

```
<h3>Introduction--h3</h3>
```

4. <h4> — Sub-sub-section Heading

The <h4> tag defines a heading inside an <h3> block. In the screenshot, the <h4> wraps the VS Code description paragraph and a <pre> block.

5. <h5> — Minor Heading

The <h5> tag is used for **smaller, less prominent headings**. In this example it announces the college info section.

```
<h5> NCIT college is located at Balkumari lalitpur <br>  
It contains many stream like be SE,CE,Civil--</h5>
```

6. <h6> — Smallest Heading

The <h6> tag is the **smallest heading** in HTML. It is typically used for fine print, footnotes, or closing remarks.

```
<h6><center>thank you--</center></h6>
```

|

thank you--h6

7. Preformatted tag <pre>

The <pre> (preformatted) tag preserves **whitespace, spaces, tabs, and line breaks** exactly as written in the HTML source. Text inside <pre> is displayed in a **monospace font**. It was used in the screenshot inside <h4> to display indented code-like text.

8. Superscripts and Subscripts

1. Superscript <sup>

- Used to show **power/exponent** (something raised above the line).
- Common in math formulas

$(a+b)^2$ <!-- means $(a+b)^2$

$(a-b)^2$ <!-- means $(a-b)^2$ -->

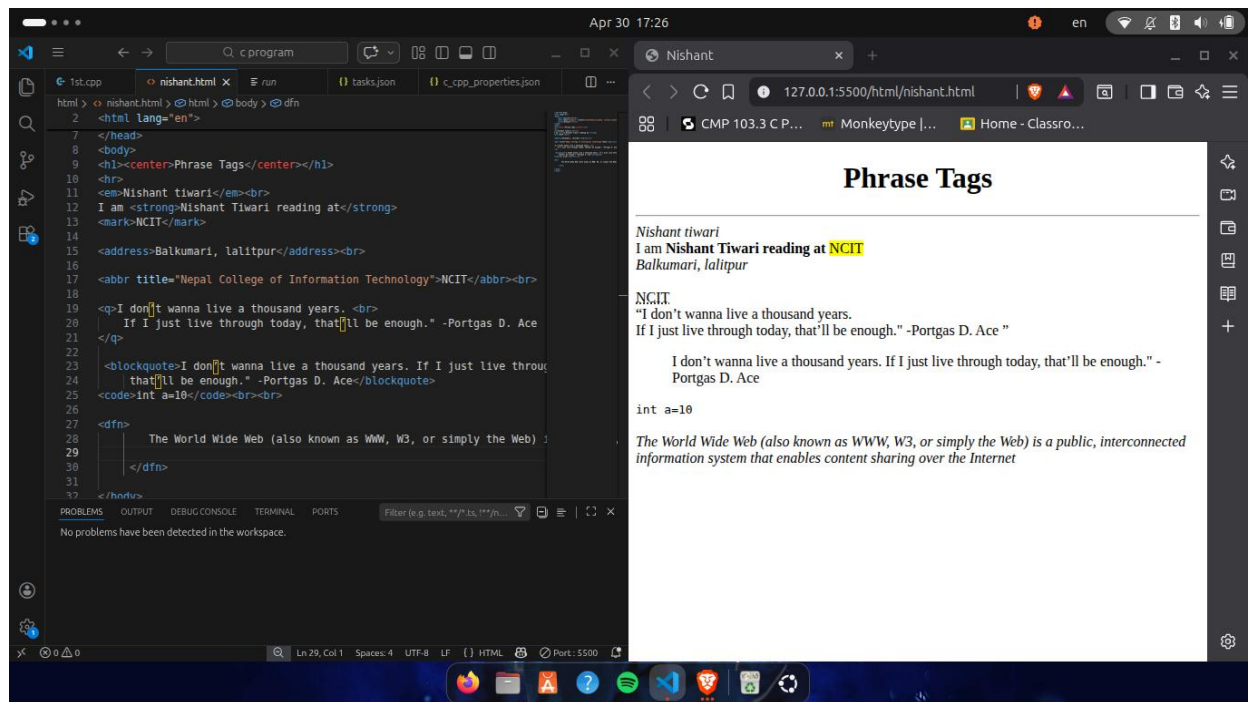
▼ 2. Subscript <sub>

- Used to show **base, index, or position** (written below the line).
- Common in binary numbers, chemistry, etc.

9. Hr line

The <hr> tag is used to create a **horizontal line** in a webpage.

TASK:PHRASE TAGS



Detailed Tag Explanations with Code

1. `<em>` — Emphasis (Italic)

The `<em>` tag marks text with *stress emphasis*. Browsers display it in italics by default.

```
<em>Nishant tiwari</em>
```

2. `<strong>` — Strong Importance (Bold)

The `<strong>` tag indicates strong importance. Browsers render it in bold.

```
I am <strong>Nishant Tiwari reading at</strong> NCIT
```

3. `<mark>` — Highlighted Text

The `<mark>` tag highlights text — typically shown with a yellow background, like a marker pen.

```
<mark>NCIT</mark>
```

4. `<code>` — Inline Code

The `<code>` tag displays a fragment of computer code using a monospace font.

```
<code>int a=10</code>
```

5. `<q>` — Short Inline Quotation

The `<q>` tag defines a short inline quotation. Browsers automatically add quotation marks around the content.

```
<q>I don't wanna live a thousand years.</q>  
<q>If I just live through today, that'll be enough.</q> -Portgas D. Ace
```

6. <blockquote> — Block Quotation

The <blockquote> tag is used for longer quotations. The content is indented by browsers and displayed as a separate block.

```
<blockquote>  
  I don't wanna live a thousand years. If I just live through today,  
  that'll be enough. -Portgas D. Ace  
</blockquote>
```

I don't wanna live a thousand years. If I just live through today, that'll be enough. -Portgas D. Ace

7. <abbr> — Abbreviation

The <abbr> tag defines an abbreviation. The title="" attribute provides the full form as a tooltip on hover.

```
<abbr title="Nepal College of Information Technology">NCIT</abbr>
```

8. <address> — Contact Information

The <address> tag provides contact information for the nearest <article> or <body> element. Browsers italicize it by default.

```
<address>Balkumari, lalitpur</address>
```

9. <dfn> — Definition Term

The <dfn> tag marks the defining instance of a term. It signals that this is where a word is being defined.

```
<dfn>  
  The World Wide Web (also known as WWW, W3, or simply the Web) is a  
  public,  
  interconnected information system that enables content sharing over the  
  Internet  
</dfn>
```

Output: *The World Wide Web...* (rendered in italics)

Task:HTML Additional Tags

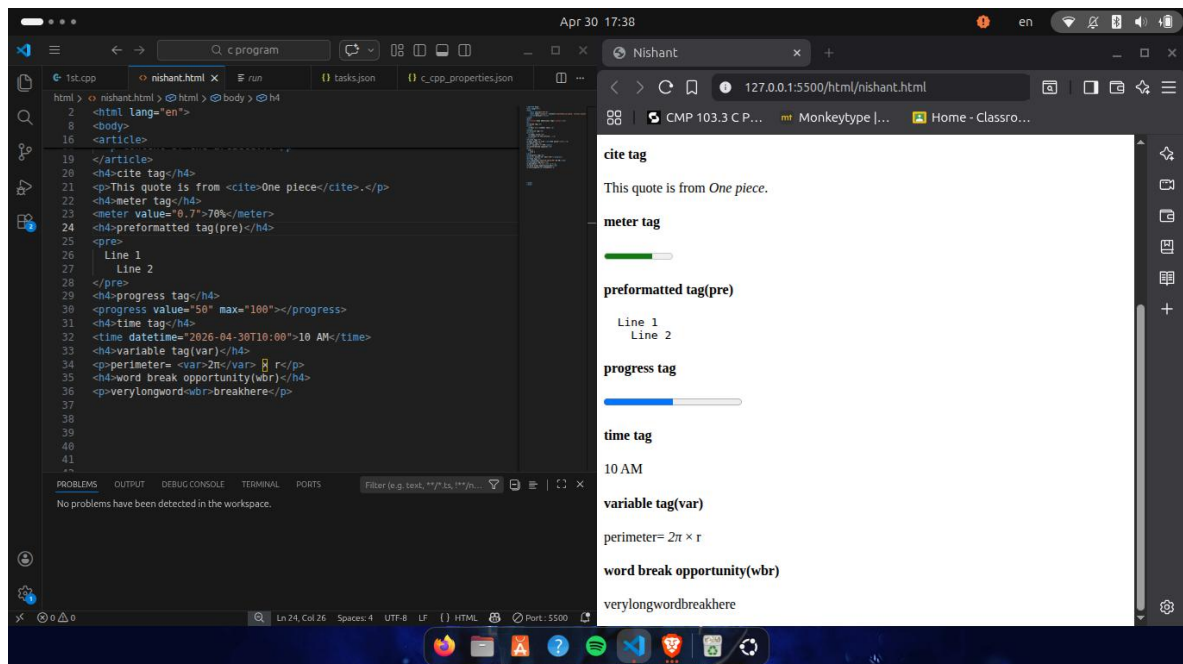
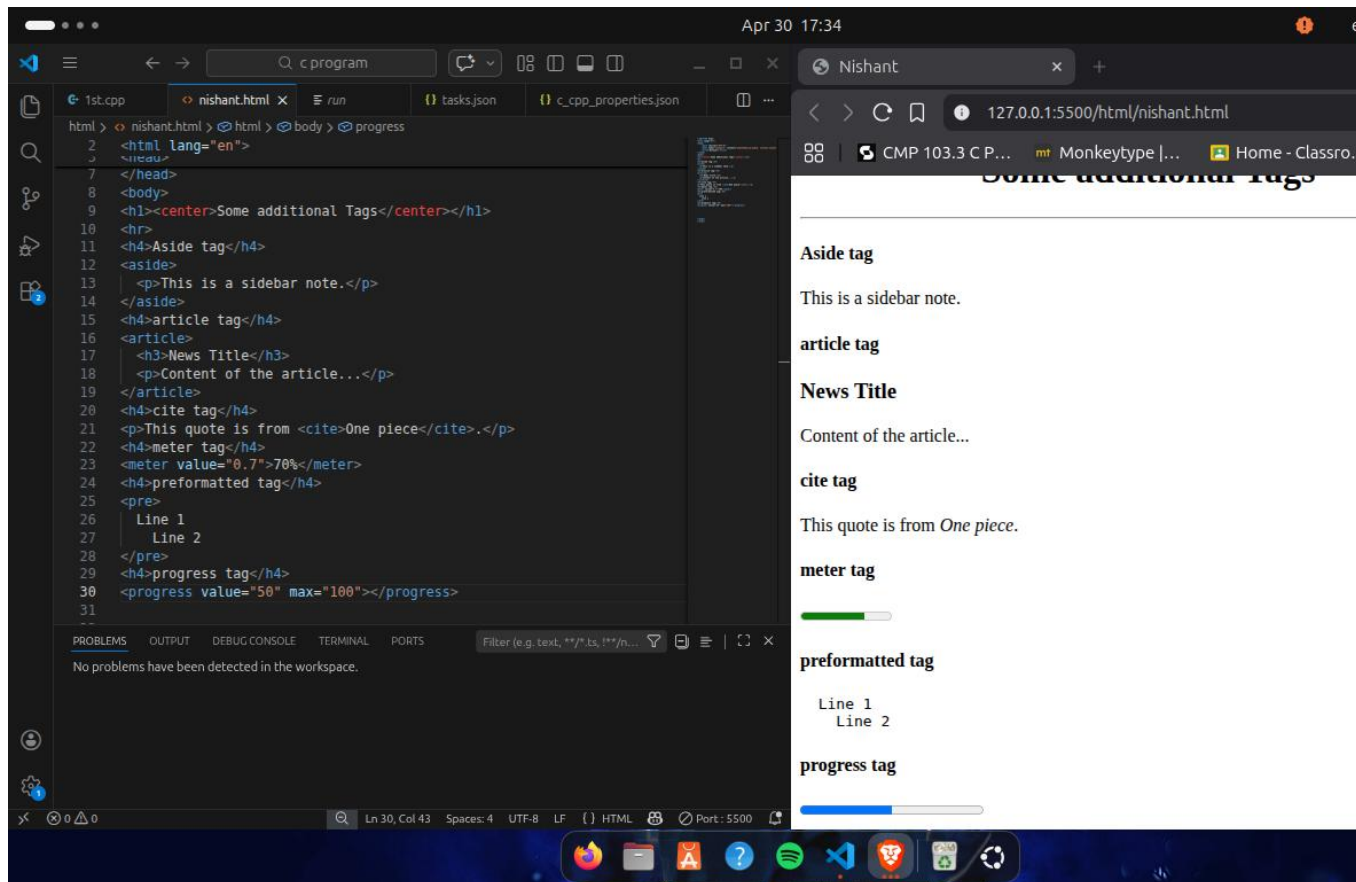


Fig:additional tag vs code

Explanation:

1. <aside> Tag

The <aside> tag defines content that is tangentially related to the main content of the page. It is typically used for sidebars, pull quotes, advertising, or other content that is considered separate from the main content.

Syntax:

```
<aside>
  <p>This is a sidebar note.</p>
</aside>
```

2. <article> Tag

The <article> tag specifies independent, self-contained content. It should make sense on its own and be distributable independently from the rest of the site (e.g., a news article, blog post, or forum post).

Syntax:

```
<article>
  <h3>News Title</h3>
  <p>Content of the article...</p>
</article>
```

3. <cite> Tag

The <cite> tag defines the title of a creative work (e.g., a book, a poem, a song, a movie, a painting, a sculpture, etc.). The text in a <cite> element is usually rendered in italic.

Syntax:

```
<p>This quote is from <cite>One piece</cite>.</p>
```

4. <meter> Tag

The <meter> tag defines a scalar measurement within a known range, or a fractional value. It is also known as a gauge. Examples: disk usage, the relevance of a query result, etc.

Syntax:

```
<meter value="0.7">70%</meter>
```

5. <pre> Tag — Preformatted Text

The <pre> tag defines preformatted text. Text in a <pre> element is displayed in a fixed-width font, and the text preserves both spaces and line breaks. The text will be displayed exactly as written in the HTML source code.

Syntax:

```
<pre>
Line 1
```

```
Line 2
</pre>
```

6. <progress> Tag

The <progress> tag represents the completion progress of a task. It is always used in combination with a JavaScript code to show the progress of a task. It displays a progress bar in the browser.

Syntax:

```
<progress value="50" max="100"></progress>
```

7. <time> Tag

The <time> tag defines a specific time (or datetime). The datetime attribute is used to translate the time into a machine-readable format so that browsers can offer to add date reminders through the user's calendar, and search engines can produce smarter search results.

Syntax:

```
<time datetime="2026-04-30T10:00">10 AM</time>
```

Browser Output:

10 AM

8. <var> Tag — Variable

The <var> tag defines a variable in programming or in a mathematical expression. The content inside is typically displayed in italic.

Syntax:

```
<p>perimeter= <var>2π</var> × r</p>
```

9. <wbr> Tag — Word Break Opportunity

The <wbr> (Word Break Opportunity) tag specifies where in a text it would be acceptable to add a line break. This is useful when you have a very long word and you want the browser to break it at a specific place when the viewport is too narrow.

Syntax:

```
<p>verylongword<wbr>breakhere</p>
```


Task1:

Write HTML code with following

Title: Your name

Body:

Heading H1: This is my 1st lab

Break

Horizontal line

Heading H2: I m writing HTML code

Break

Heading H3:

Name: Your name

Faculty: Your Faculty

Roll no: Your Roll Number

Paragraph p : Write about yourself minimum 4 lines

Save it as Yourname.html and run it from browser

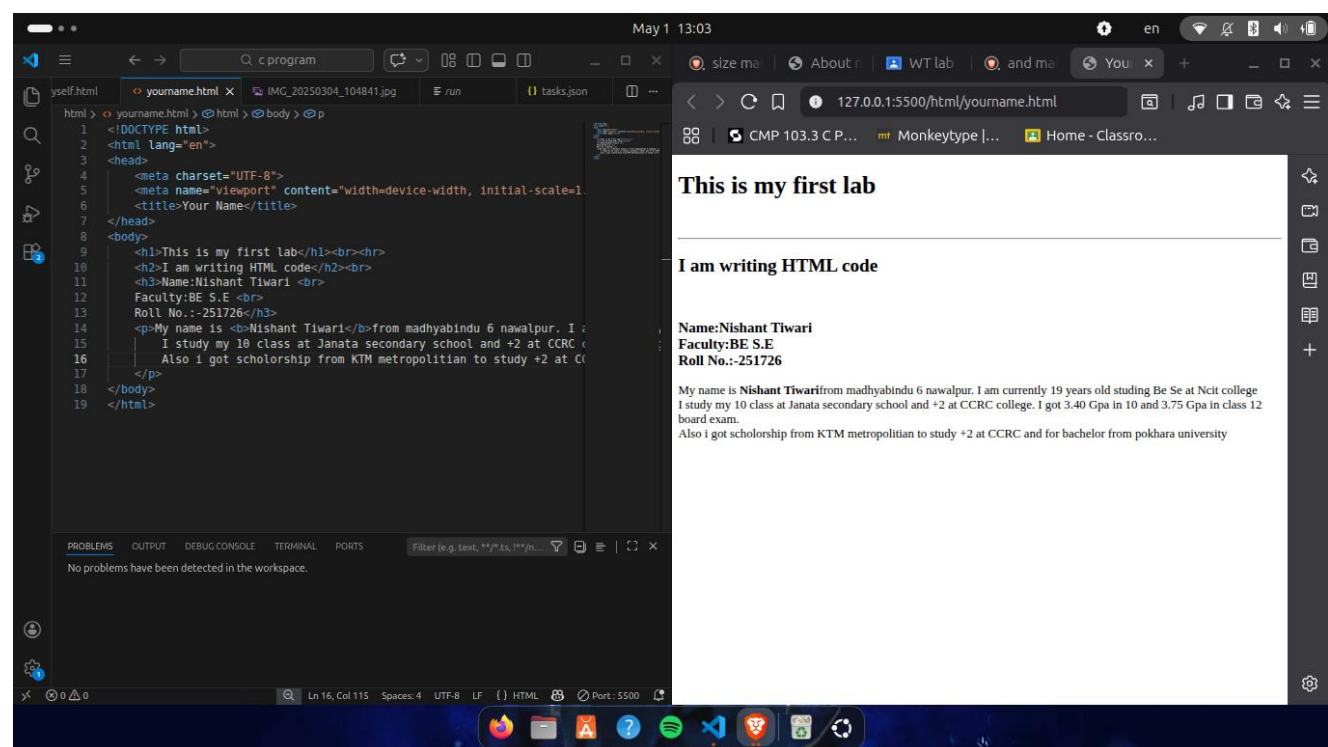
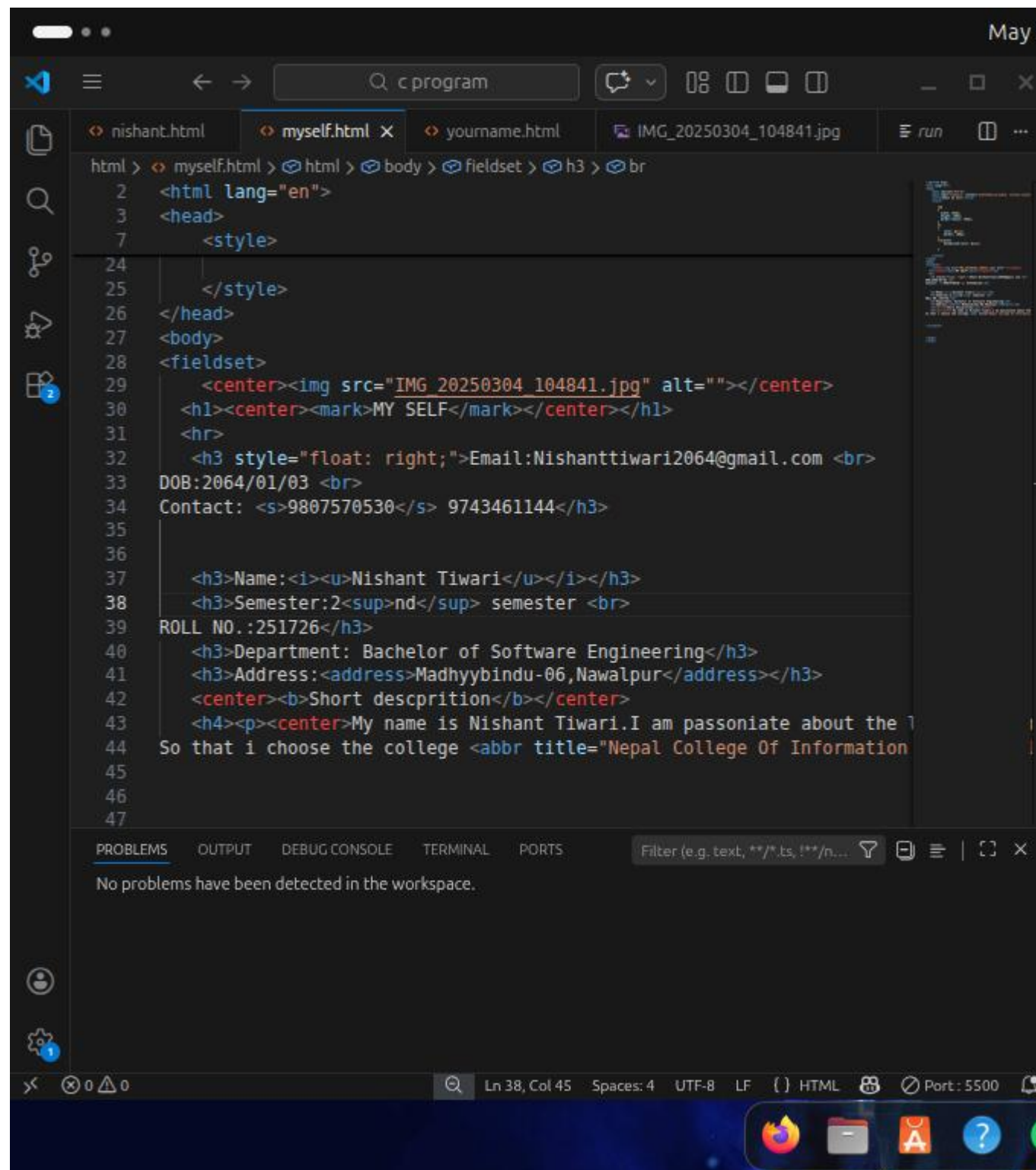


Fig:task 1

Task 2 and 3: Sample



The image shows a screenshot of the Visual Studio Code editor interface. The top bar displays the date 'May' and the search bar contains 'c program'. The file explorer on the left shows the project structure with files 'nishant.html', 'myself.html', 'yourname.html', and 'IMG_20250304_104841.jpg'. The 'myself.html' file is open in the editor, showing the following HTML code:

```
html > myself.html > html > body > fieldset > h3 > br
2  <html lang="en">
3  <head>
7    <style>
24
25  </style>
26 </head>
27 <body>
28 <fieldset>
29   <center></center>
30   <h1><center><mark>MY SELF</mark></center></h1>
31   <hr>
32   <h3 style="float: right;">Email:Nishanttiwari2064@gmail.com <br>
33   DOB:2064/01/03 <br>
34   Contact: <s>9807570530</s> 9743461144</h3>
35
36
37   <h3>Name:<i><u>Nishant Tiwari</u></i></h3>
38   <h3>Semester:2<sup>nd</sup> semester <br>
39   ROLL NO.:251726</h3>
40   <h3>Department: Bachelor of Software Engineering</h3>
41   <h3>Address:<address>Madhybindu-06,Nawalpur</address></h3>
42   <center><b>Short descprition</b></center>
43   <h4><p><center>My name is Nishant Tiwari.I am passoniante about the 1
44   So that i choose the college <abbr title="Nepal College Of Information
45
46
47
```

The bottom panel shows the 'PROBLEMS' tab, indicating 'No problems have been detected in the workspace.' The status bar at the bottom shows 'Ln 38, Col 45', 'Spaces: 4', 'UTF-8', 'LF', 'HTML', and 'Port: 5500'.

Fig:source code

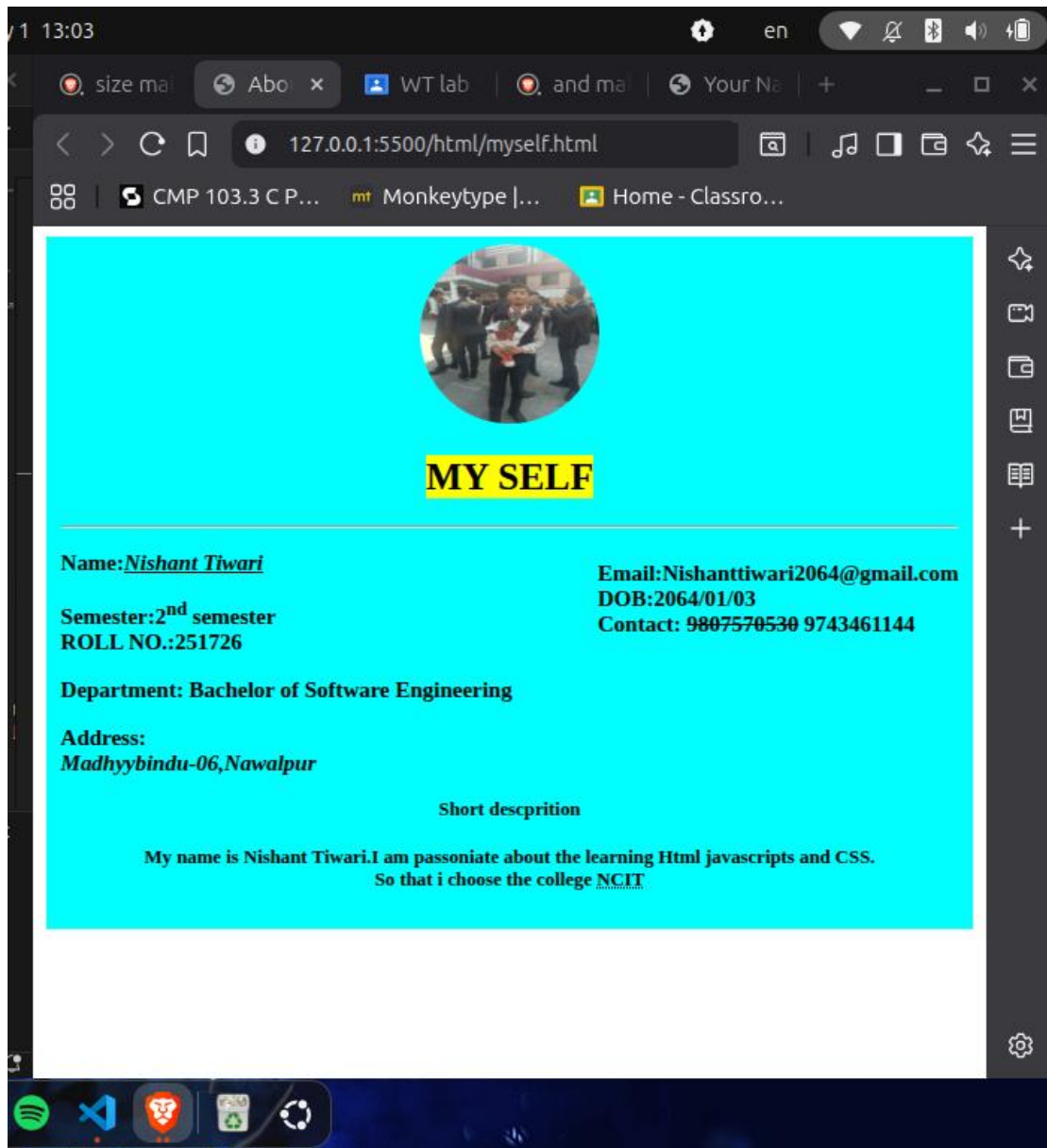


Fig:output